



## AQ4.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

**Level 1**

Find the rank of the following matrix:  $\begin{bmatrix} 0 & 1 & 3 & 1 \\ 1 & 0 & 0 & 1 \\ 3 & 2 & 0 & 3 \end{bmatrix}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

Consider the following set of vectors  $SS$  in  $R^3$  to answer the questions 2, 3 and 4.

$S = \{(1, 2, 0), (0, 3, 1), (3, 3, -1), (3, 0, -2)\}$

1 point

Let  $V$  be the vector space spanned by the vectors  $(1, 2, 0)$  and  $(0, 3, 1)$ . Which one of the following is correct?

☐

$V = \{(a, 2a + 3b, b) \mid a, b \in R\}$

☐

$V = \{(0, 2a - 3b, b) \mid a, b \in R\}$

☐

$V = \{(-a, 2a + 3b, 0) \mid a, b \in R\}$

☐

$V = \{(0, 2a + 3b, 0) \mid a, b \in R\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$V = \{(a, 2a + 3b, b) \mid a, b \in R\}$

If the vectors in  $SS$  are written as the columns of a matrix  $A$ , then what will be the rank of  $A$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

What will the dimension of the vector space spanned by  $SS$  be?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point



1 point

Choose the set of correct options from the following.

- ☐ Row rank and column rank of a matrix is always the same.
- ☐ The rank of a zero matrix is always 0.
- ☐ The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.
- ☐

Rank of the  $n \times n$  identity matrix is 1 for any  $n \in \mathbb{N}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Row rank and column rank of a matrix is always the same.

The rank of a zero matrix is always 0.

The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.

## Level 2

1 point

Consider the following upper triangular matrix to choose the correct options.

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

where  $a, b, c, d, e, f \in \mathbb{R}$ .

☐ If  $f = 0$ , then the rank of the matrix must be less than or equal to 2.

☐ If  $f = 0$ , then the rank of the matrix must be exactly 2.

☐ If  $a, b, c, d, e, f$  are all non-zero then the rank of the matrix must be 3.

☐ If  $a, d, f$  are non-zero then the rank of the matrix must be 3.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $f = 0$ , then the rank of the matrix must be less than or equal to 2.

If  $a, b, c, d, e, f$  are all non-zero then the rank of the matrix must be 3.

If  $a, d, f$  are non-zero then the rank of the matrix must be 3.

If rank of the matrix  $\begin{bmatrix} 2 & -3 & 4 \\ 0 & 1 & -2 \\ 1 & -3 & a \end{bmatrix}$  is 2 then find the value of  $a$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 5

1 point

Find the rank of the matrix  $A$ , where  $A = [a_{ij}]$  is of order  $3 \times 3$  and  $a_{i,j} = \min\{i, j\}$ ,  $i, j = 1, 2, 3$ .

[Hint: Write down the matrix and apply row operations to derive its reduced row echelon form.]

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

What is the rank of the matrix  $A = \begin{bmatrix} x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$ , where  $x, y, z$  are distinct non-zero integers?

non-zero integers?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

Rank of a  $3 \times 3$  matrix  $A$  is 3 and another  $3 \times 3$  matrix  $B$  is 2. What is the minimum possible value for the rank of  $ABAB$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

[Check Answers](#)

Your score is: 0/10